BUILD

WITH

भारत 2.0

NATIONAL LEVEL HACKATHON

TEAM NAME: Team Sustainovators

PROBLEM STATEMENT: Unsafe Drinking Water in Rural Households Near Mining Belts

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PROBLEM STATEMENT

“Rural households near Jharkhand's mining belts drink contaminated water blindly — with no detection, and no purification designed for their specific water problem.”

- Households in mining-affected districts of Jharkhand (e.g., Ramgarh, Dhanbad, West Singhbhum) draw drinking water directly from wells, hand pumps, and borewells — with no treatment at the point of use.
- Water in these areas commonly carries:
  - Suspended silt & coal dust from mining runoff
  - Acidic pH from acid mine drainage
  - Dissolved heavy metals (iron, lead, copper) from mining activity
  - Fluoride/arsenic, a documented issue in parts of Jharkhand's groundwater, unrelated to mining but present in the same rural belt
- No household-level way exists to check, in real time, what's actually in the water before drinking it.
- Existing purifiers (home RO units) are generic, need constant electricity, and aren't built for the specific contamination profile of mining-belt groundwater.

SOLUTION

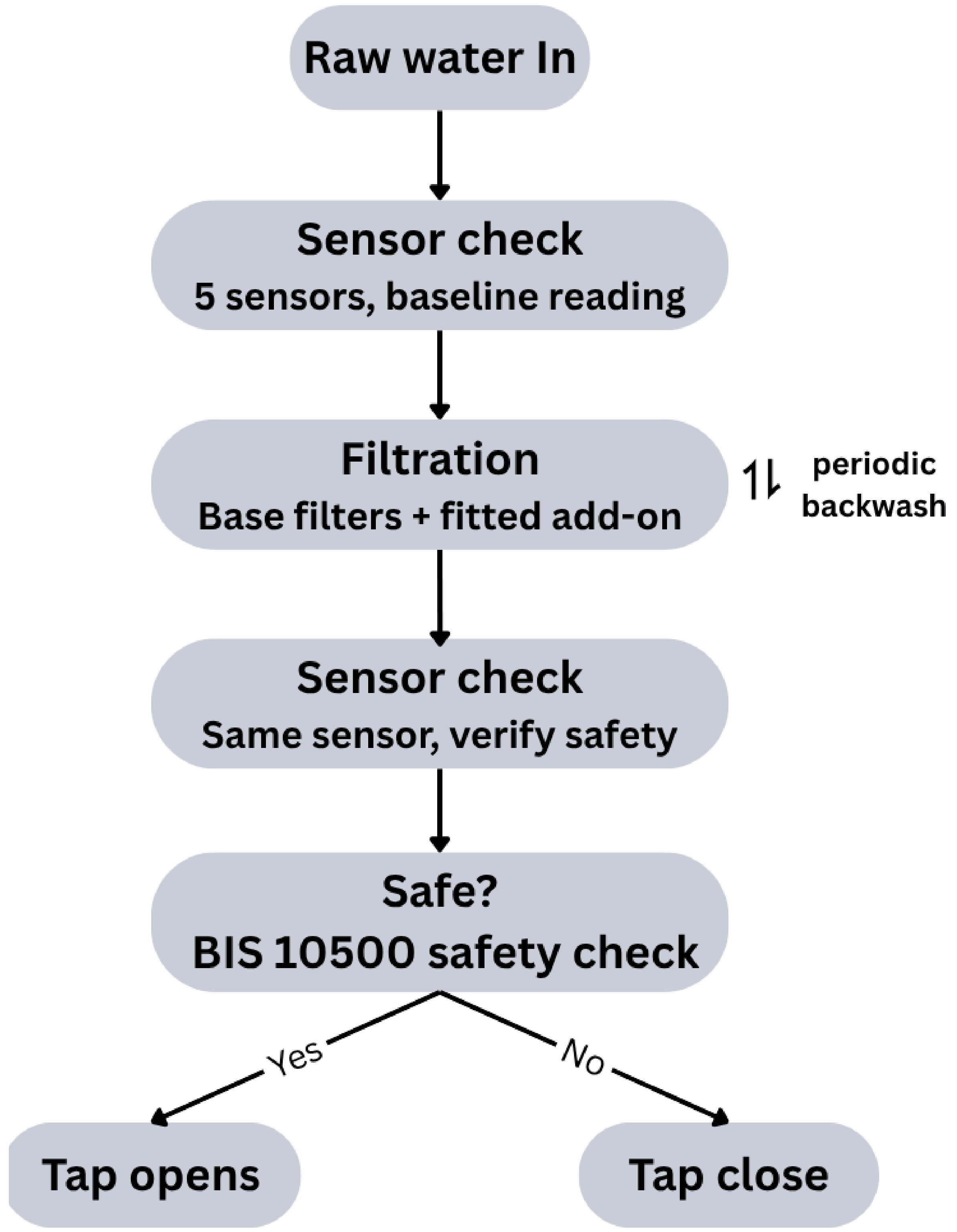
“A solar-powered, self-cleaning water purifier that's fitted to each home's actual water problem, and only lets safe water through.”

- A solar-powered water purifier installed right at the household's water outlet — cleans water and checks its safety before it reaches the tap.
- Customizable per home: every unit comes with a base filter for mud, smell, and germs — plus extra add-on filters (for fluoride, heavy metals, or high dissolved salts) fitted only in homes that actually need them, based on a local water test.
- Checks the water twice — once before cleaning, once after — using the same set of sensors, so the comparison is accurate and fair.
- Self-cleaning filters: the system periodically flushes itself out, so filters last longer and don't need frequent replacement.
- Locks the tap automatically if the cleaned water doesn't meet safety standards — water only flows when it's actually safe.
- Works fully off-grid, with no need for constant electricity or internet.

FLOW OF SOLUTION

1. Raw water enters from the home's tap/borewell line.
2. Sensors take a baseline reading (pH, turbidity, TDS, temperature, spectral scan).
3. Water passes through base filters + any add-on modules fitted for that home.
4. The same sensors check the water again after filtration.
5. If it passes safety thresholds → tap opens, water flows.
6. If it fails → tap stays locked, alert triggers.
7. Filters are periodically backwashed to stay clog-free.

- Note: BIS 10500 is the Indian government's official safety standard for drinking water — this is the benchmark our sensors check against.



TECH STACK

Sensing

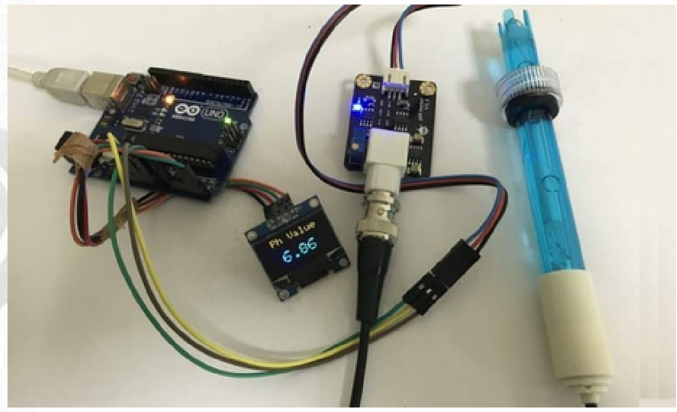
- pH probe — measures acidity/alkalinity
- Turbidity sensor — measures cloudiness/silt
- TDS/conductivity probe — measures dissolved solids
- Temperature sensor
- AS7262 spectral sensor — detects which metal is likely present (iron, lead, copper)

Processing & Control

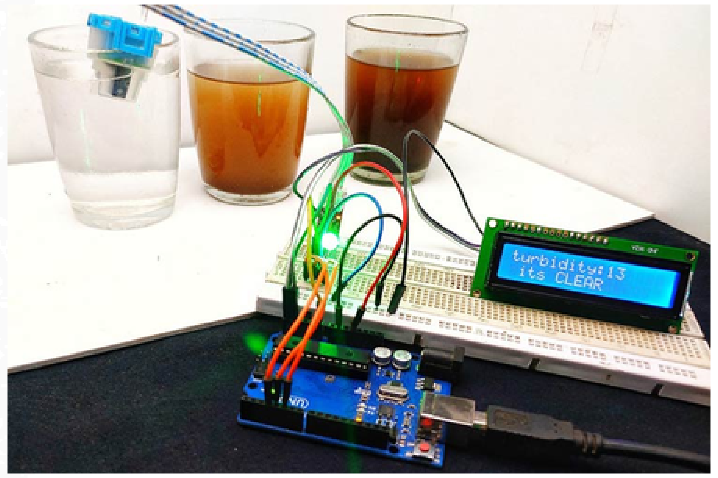
- ESP32 microcontroller — reads all sensors, runs decision logic, controls valves
- Ardiuno uno – used in testing of sensors
- Rule-based threshold logic — calibrated against known reference samples, no AI training needed

Display & Monitoring

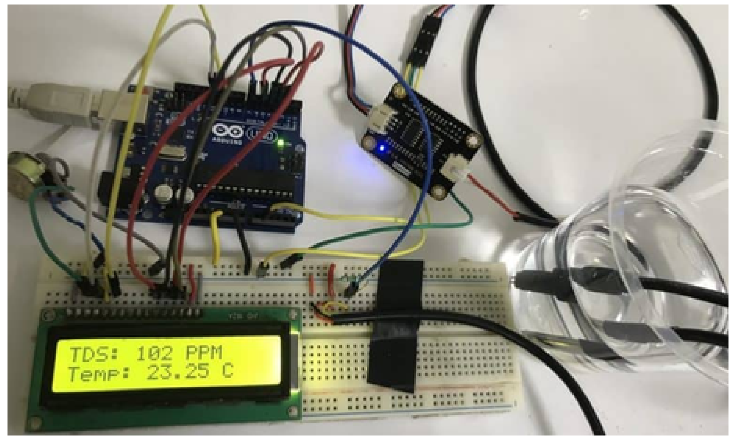
- Serial Monitor (via USB) — used during development/demo to view live sensor readings
- 16x2 LCD / OLED display — shows real-time readings and safety status locally on the deployed unit, no phone or laptop needed



TESTING OF PH SENSOR



TESTING OF TURBDITY SENSOR



TESTING OF TEMPERATURE AND TDS PROBE

TECH STACK

Filtration & Actuation

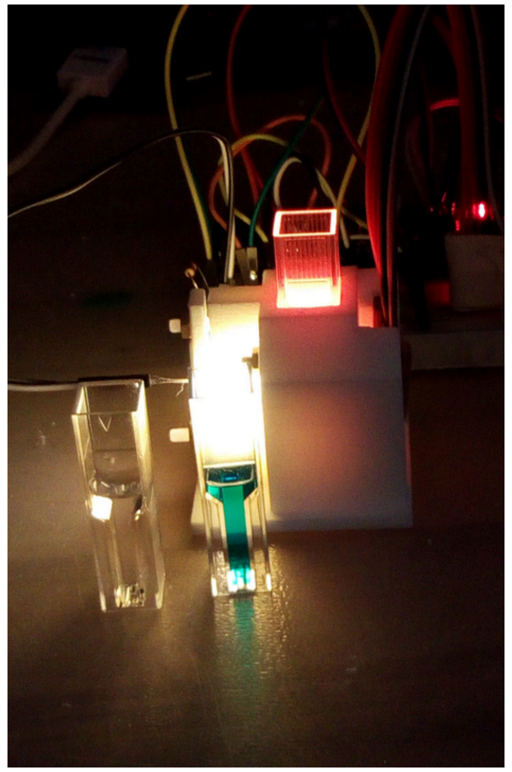
- Base filters: sediment filter, activated carbon/resin, UV-C disinfection
- Optional add-on modules: activated alumina (fluoride/arsenic), KDF-55 resin (heavy metals), RO membrane + small pump
- Solenoid valves: tap lock, sensor diverter (shared before/after readings), backwash reversing valve

Power System

- 4x 3.7V 2200mAh Li-ion cells, wired in series (4S1P) — 14.8V nominal, 32.5Wh total
- 4S BMS (Battery Management System) — mandatory safety circuit, prevents overcharge/over-discharge
- 12V buck converter — regulates fluctuating pack voltage to a stable 12V for solenoids/UV-C/pump
- 5V/3.3V buck converters — power ESP32 and sensors
- Solar panel (20W) + lithium-compatible solar charge module — off-grid daily recharge

Alerts

- Buzzer — local alert
- Optional GSM/SMS module — remote alert, no internet required



TESTING OF AS7262 SPECTRAL SENSOR

BILL OF MATERIALS

Sensing and display	
ESP32 DevKit v1	450
pH sensor module	900
Turbidity sensor	350
TDS sensor module	400
Waterproof temp sensor	120
AS7262 spectral sensor	750
16x2 LCD display	250

Power system	
Li-ion cells 3.7V 2200mAh (x4)	700
4S BMS module	200
Buck converter, 12V stage	120
Buck converters, 5V/3.3V (x2)	100
Solar panel 12V 20W	850
Lithium solar charge module	300
Buzzer module	30
GSM module (optional)	550

Filtration and valves	
Sediment cartridge + housing	450
Carbon cartridge + housing	650
UV-C LED module	1,100
Solenoid, tap lock	350
Solenoid, sensor diverter	450
Solenoid, backwash reverse	350
Relay modules (x3)	180
MOSFET driver module	70

Plumbing, wiring and enclosure	
Enclosure box	300
Raw water container	250
Clean water container	250
Backwash drain bucket	100
Silicone tubing (4m)	200
Push-fit connectors (~12)	300
Jumper wires (3 packs)	180
Hookup wire 22 AWG (5m)	100
Perfboard	80
Misc: screws, ties, solder	250

Total Cost ~= Rs.12,000/-

UNIQUE SELLING PROPOSITION (USP)

- **Identifies the actual contaminant** – spectral sensor detects iron, lead, or copper specifically, not just a generic "unsafe" flag.
- **Modular per household** – base unit + only the add-on filters a home actually needs, based on real water testing.
- **Accurate before/after comparison** – one shared sensor set for both readings, not two separate ones.
- **Self-cleaning filters** – automatic backwash extends filter life, cuts maintenance cost.
- **Fully off-grid** – solar-powered, no electricity or internet dependency.
- **Hardware-locked safety** – tap physically cannot open on unsafe water.

Water Treatment comes nearly into the health sector , In health sector AI predictions may produce unnecessary calls which are not tolerable...so we use a special algorithm to determine the quality of the water which has near to 100% accuracy during the testing which is a major addition to our USP.

ITS NOT A PREDICTION, ITS A CONFIRMATION!

FEASIBILITY & COMPETITORS

Feasibility

- ₹12,000/household – off-the-shelf parts, no lab equipment, no pump needed

Competitors

- Water ATMs (Sarvajal): need a separate plant + card system – not household-level
- Home RO purifiers: no contamination detection, same setup for every water problem

Can the Government Afford It? – Example: West Singhbhum

- Jharkhand's per capita income: ₹1,14,271/year – our unit costs about one month's income, one-time, not recurring.
- West Singhbhum already holds ₹600 crore in DMF funds – money earmarked specifically for mining-affected community welfare, currently under-spent on water.
- That fund alone could cover ~5 lakh units – enough for every household in its mining-affected blocks, several times over.

“***No new government spending needed – just redirecting existing, earmarked mining-welfare funds to the exact problem they exist for***”.

RESEARCH & REFERENCE

- "Removal of heavy metal ions from wastewater using ion exchange resin in a batch process," ScienceDirect, 2024. Link : https://www.sciencedirect.com/science/article/pii/S1026918524000490
- "Risk assessment due to intake of metals in groundwater of East Bokaro Coalfield, Jharkhand," Link : https://link.springer.com/article/10.1007/s12403-016-0201-2
- Springer study on Ramgarh district's Kuju coal mining area (2024–25) found elevated TDS, calcium, magnesium, and sulphate directly linked to mining activity. Link: https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/5078889
- "Evaluation of the Surface Water Quality Index of Jharia Coal Mining Region," Springer. Link : https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/5078889
- U.S. patent literature confirms plain activated carbon "lacks usefulness for most heavy metals" and needs to be paired with metal-specific media. Link:https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/5078889

- **One-line takeaway**: Both halves of our design are backed by published research — that filter media must be matched to the specific contaminant, and that Jharkhand's mining-belt groundwater is measurably unsafe without treatment.
- Github Repo Link:

THANK YOU